AERO40008 – Structures

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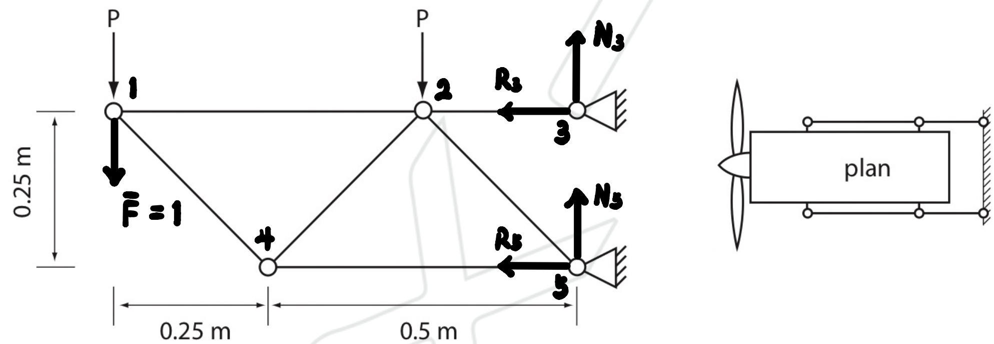
Virtual Work

(**1.**) Use virtual work to determine the vertical deflection of the node furthest from the supports in the engine mount in Q. 7 from Tutorial Sheet 1.

One side of a small engine mount is represented by the truss shown in the figure. The engine weighs 250 kg and is attached to each support at two points where it transmits equal force P. The bars are made of Aluminium Alloy (E = 72 GPa). Determine the required cross-sectional areas of the bars if the maximum absolute value of the stress is to be kept below 200 MPa.



From tutoriul 1 (7),

$T_{12} = P$, $A_{max} = 3.066 \times 10^{-6}\, m^2$, $e = 1.37 \times 10^{-3}\, m$

$T_{14} = -\sqrt{2}P$, $A_{max} = 4.335 \times 10^{-6}\, m^2$, $e = -0.94 \times 10^{-3}\, m$

$T_{24} = \sqrt{2}P$, $A_{max} = 4.335 \times 10^{-6}\, m^2$, $e = 0.94 \times 10^{-3}\, m$

$T_{23} = 4P$, $A_{max} = 1.226 \times 10^{-6}\, m^2$, $e = 0.64 \times 10^{-3}\, m$

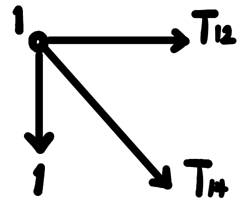
$T_{25} = -2\sqrt{2}P$, $A_{max} = 8.671 \times 10^{-6}\, m^2$, $e = -0.98 \times 10^{-3}\, m$

$T_{45} = -2P$, $A_{max} = 6.131 \times 10^{-6}\, m^2$, $e = -1.39 \times 10^{-3}\, m$

For $\bar{F} = 1$,

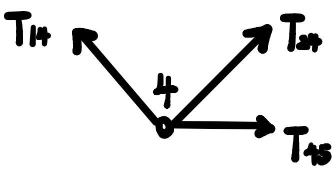
$(\rightarrow)\ R_3 + R_5 = 0 \qquad (\uparrow)\ N_3 + N_5 = 1$

$M(3),\ 0.25 \cdot R_5 = 0.75 \cdot 1 \qquad \therefore R_5 = 3 \qquad \therefore R_3 = -3$



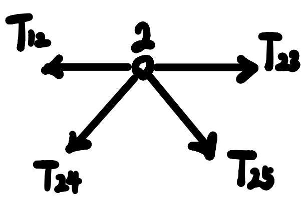
$(\rightarrow)\ T_{12} + \frac{\sqrt{2}}{2} T_{14} = 0$

$(\uparrow)\ \frac{\sqrt{2}}{2} T_{14} + 1 = 0 \qquad \therefore T_{14} = -\sqrt{2} \qquad \therefore T_{12} = 1$



$(\uparrow)\ \frac{\sqrt{2}}{2} T_{14} + \frac{\sqrt{2}}{2} T_{24} = 0 \qquad \therefore T_{24} = -T_{14} = \sqrt{2}$

$(\rightarrow)\ \frac{\sqrt{2}}{2} T_{14} = \frac{\sqrt{2}}{2} T_{24} + T_{45} \qquad \therefore T_{45} = -2$



$(\uparrow)\ \frac{\sqrt{2}}{2} T_{24} + \frac{\sqrt{2}}{2} T_{25} = 0 \qquad \therefore T_{25} = -T_{24} = -\sqrt{2}$

$(\rightarrow)\ T_{12} + \frac{\sqrt{2}}{2} T_{24} = T_{23} + \frac{\sqrt{2}}{2} T_{25} \qquad \therefore T_{23} = 3$

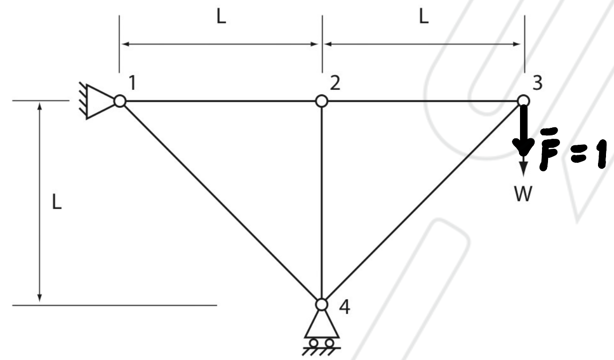
$\therefore\ 1 \cdot v_1 = 1 \cdot e_{12} - \sqrt{2} \cdot e_{14} + \sqrt{2} \cdot e_{24} + 3 \cdot e_{23} - \sqrt{2} \cdot e_{25} - 2 \cdot e_{45}$

$\therefore\ v_1 = 10.41 \times 10^{-3}\ m$

$= 10.41\ mm$

(**2.**) Determine the vertical displacement of joint 3 in the the structure in Q. 5 from Tutorial Sheet 2 using virtual work. If bar 1-4 of the framework in this question were heated by ΔT show that the vertical deflection of joint 3 due to this heating would be $2\alpha L\Delta T$ (upwards).

In the structure shown in the figure, all members have cross-section area A, and Young's modulus E. Determine the joint displacements under the loading shown, and sketch the deformed shape.



From tutorial 2 (5),

$$e_{12} = e_{23} = \frac{WL}{AE}$$

$$e_{14} = e_{34} = -\frac{2WL}{AE}$$

$$e_{24} = 0$$

For $\bar{F} = 1$, as shown in tutorial (5)

$$T_{12} = T_{23} = 1$$

$$T_{14} = T_{34} = -\sqrt{2}$$

$$T_{24} = 0$$

$\therefore 1 \cdot v_3 = 1 \cdot e_{12} + 1 \cdot e_{23} - \sqrt{2} \cdot e_{14} - \sqrt{2} \cdot e_{34}$

$\therefore v_3 = (4\sqrt{2} + 2) \dfrac{WL}{AE}$

By heating.

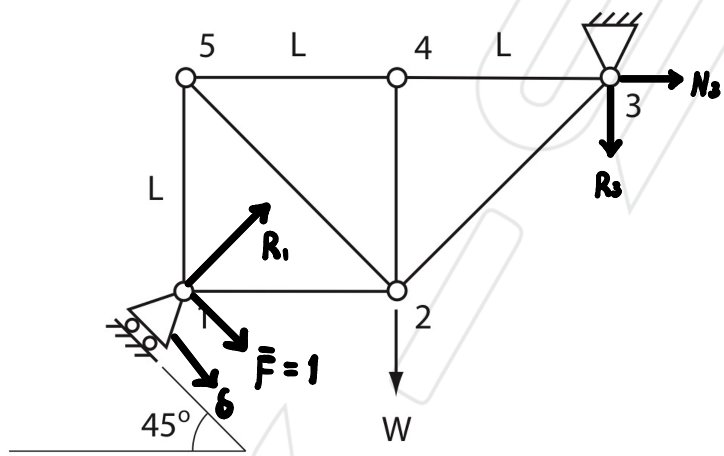
$\varepsilon_{14} = \alpha \Delta T$

$\Delta e_{14} = \varepsilon_{14} L_{14} = \sqrt{2} \alpha L \Delta T$

$\therefore \Delta v_3 = -\sqrt{2} \cdot \Delta e_{14} = -2\alpha L \Delta T$ (downwards)

$\therefore \Delta v_3 = 2\alpha L \Delta T$ (upwards)

(**3.**) The pin-jointed framework shown in the figure consists of bars which all have the same cross-section area A and Young's Modulus E. If the support at joint 1 is constrained to move on the incline shown, calculate its displacement due to the load W.



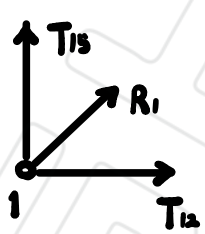
$$M = b + r - 2j = 7 + 3 - 2 \times 5 = 0$$

By inspection, statically determine

$(\rightarrow)\ \frac{\sqrt{2}}{2} R_1 + N_3 = 0 \qquad (\uparrow)\ \frac{\sqrt{2}}{2} R_1 = W + R_3$

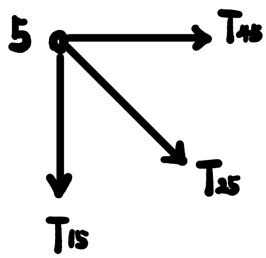
$M(2),\ \frac{\sqrt{2}}{2} L \cdot R_1 + L \cdot R_3 + L \cdot N_3 = 0 \qquad \therefore R_3 = 0$

$\therefore R_1 = \sqrt{2} W \qquad \therefore N_3 = -W$



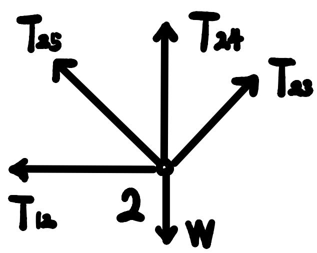
$(\rightarrow)\ \frac{\sqrt{2}}{2} R_1 + T_{12} = 0 \qquad \therefore T_{12} = -W$

$(\uparrow)\ \frac{\sqrt{2}}{2} R_1 + T_{15} = 0 \qquad \therefore T_{15} = -W$



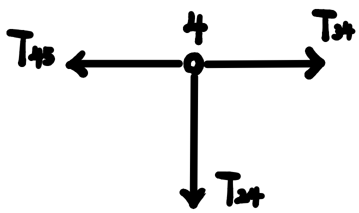
$(\rightarrow)\quad T_{45} + \frac{\sqrt{2}}{2} T_{25} = 0$

$(\uparrow)\quad T_{15} + \frac{\sqrt{2}}{2} T_{25} = 0 \qquad \therefore T_{25} = \sqrt{2}W \qquad \therefore T_{45} = -W$



$(\rightarrow)\quad T_{12} + \frac{\sqrt{2}}{2} T_{25} = \frac{\sqrt{2}}{2} T_{23} \qquad \therefore T_{23} = 0$

$(\uparrow)\quad T_{24} + \frac{\sqrt{2}}{2} T_{25} + \frac{\sqrt{2}}{2} T_{23} = W \qquad T_{24} = 0$



$(\rightarrow)\quad T_{34} = T_{45} \qquad \therefore T_{34} = -W$

$(\uparrow)\quad T_{24} = 0$

$$\therefore e_{12} = e_{15} = e_{45} = e_{34} = -\frac{WL}{AE}$$

$$e_{25} = \frac{\sqrt{2}W\sqrt{2}L}{AE} = \frac{2WL}{AE}$$

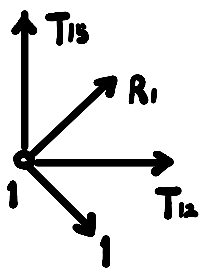
$$e_{23} = e_{24} = 0$$

For $\bar{F}_1 = 1$,

$(\rightarrow)\quad \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} R_1 + N_3 = 0 \qquad (\uparrow)\quad \frac{\sqrt{2}}{2} R_1 = R_3 + \frac{\sqrt{2}}{2}$

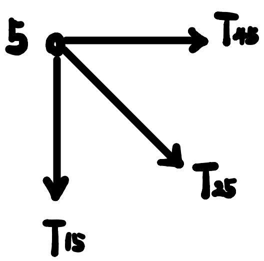
$M(1),\ 2L\cdot R_3 + L\cdot N_3 = 0 \qquad \therefore N_3 = -2R_3$

$\therefore R_3 = \sqrt{2} \qquad \therefore N_3 = -2\sqrt{2} \qquad \therefore R_1 = 3$



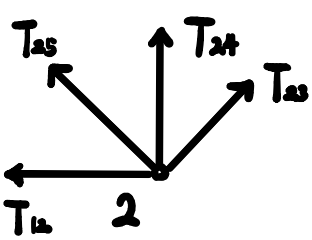
$(\rightarrow)\ \frac{\sqrt{2}}{2}R_1 + \frac{\sqrt{2}}{2} + T_{12} = 0 \qquad \therefore T_{12} = -2\sqrt{2}$

$(\uparrow)\ \frac{\sqrt{2}}{2}R_1 + T_{15} = \frac{\sqrt{2}}{2} \qquad \therefore T_{15} = -\sqrt{2}$



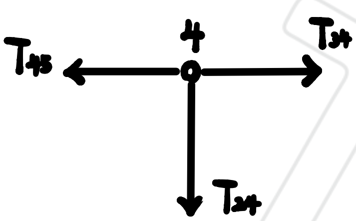
$(\rightarrow)\ T_{45} + \frac{\sqrt{2}}{2}T_{25} = 0$

$(\uparrow)\ T_{15} + \frac{\sqrt{2}}{2}T_{25} = 0 \qquad \therefore T_{25} = 2 \qquad \therefore T_{45} = -\sqrt{2}$



$(\rightarrow)\ T_{12} + \frac{\sqrt{2}}{2}T_{25} = \frac{\sqrt{2}}{2}T_{23} \qquad \therefore T_{23} = -2$

$(\uparrow)\ T_{24} + \frac{\sqrt{2}}{2}T_{25} + \frac{\sqrt{2}}{2}T_{23} = 0 \qquad T_{24} = 0$



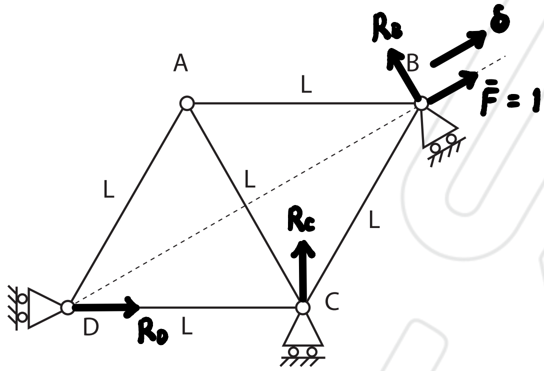
$(\rightarrow)\ T_{34} = T_{45} \qquad \therefore T_{34} = -\sqrt{2}$

$(\uparrow)\ T_{24} = 0$

$\therefore 1\cdot\delta = e_{12}T_{12} + e_{15}T_{15} + e_{45}T_{45} + e_{34}T_{34} + e_{25}T_{25}$

$\therefore \delta = (4+5\sqrt{2})\frac{WL}{AE}$

(**4.**) The figure shows a truss made from members of equal length L. Joints B, C, and D are constrained to move in the directions shown by means of roller supports. Joint B can move in the direction DB (indicated by the dotted line), joint C may move horizontally, and joint D vertically. Find the displacement of B when bar CD extends by amount e.



$$M = b + r - 2j = 5 + 3 - 2 \times 4 = 0$$

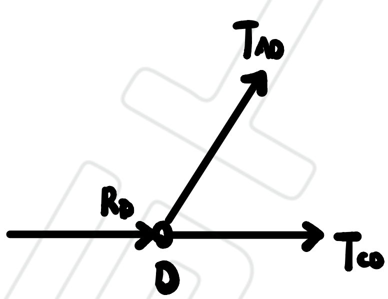
By inspection, statically determine

For $\bar{F} = 1$,

$(\rightarrow)$ $R_D + \frac{\sqrt{3}}{2} = \frac{1}{2} R_B$ $\quad$ $(\uparrow)$ $R_C + \frac{\sqrt{3}}{2} R_B + \frac{1}{2} = 0$

$M(B)$, $\frac{1}{2} L \cdot R_C = \frac{\sqrt{3}}{2} L \cdot R_D$ $\quad \therefore R_C = \sqrt{3} R_D$

$\therefore R_D = -\frac{\sqrt{3}}{3}$ $\quad \therefore R_C = -1$ $\quad \therefore R_B = \frac{\sqrt{3}}{3}$

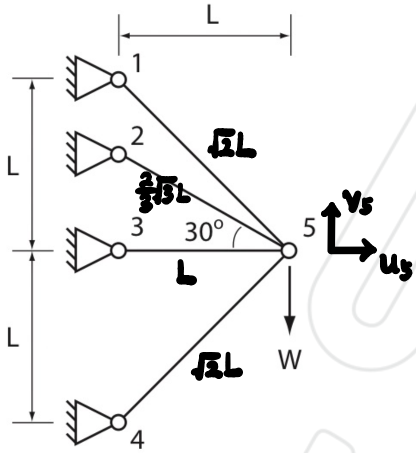


$(\uparrow)$ $\frac{\sqrt{3}}{2} T_{AD} = 0$ $\quad \therefore T_{AD} = 0$

$(\rightarrow)$ $T_{CD} + R_D + \frac{1}{2} T_{AD} = 0$ $\quad \therefore T_{CD} = \frac{\sqrt{3}}{3}$

$\therefore 1 \cdot \delta = T_{CD} \cdot e_{CD}$ $\quad \therefore \delta = \frac{\sqrt{3}}{3} e$

Redundant Frameworks



(**5.**) All the bars in the framework shown have the same cross-section area A and Young's Modulus E. Show using the procedure outlined in the lectures for statically-indeterminate structures that the bar forces are

$$\begin{aligned} T_{15} &= 0.4867W \\ T_{25} &= 0.3631W \\ T_{35} &= -0.1842W \\ T_{45} &= -0.6695W \end{aligned} \tag{1}$$

The horizontal and vertical displacements of joint 5 should be $-0.184\frac{WL}{AE}$ and $-1.158\frac{WL}{AE}$ respectively.

$M = b + r - 2j = 4 + 8 - 2 \times 5 = 2$

$\therefore$ statically indetermine with 2 redundencies

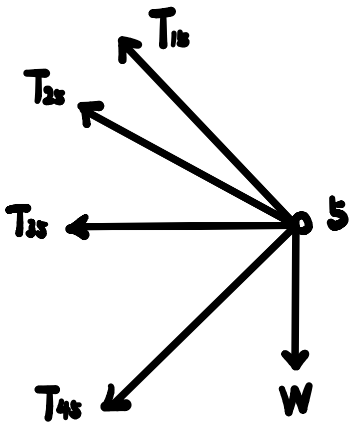
$e_{15} = \frac{\sqrt{2}}{2}(u_5 - v_5)$, $e_{25} = \frac{\sqrt{3}}{2}u_5 - \frac{1}{2}v_5$, $e_{35} = u_5$, $e_{45} = \frac{\sqrt{2}}{2}(u_5 + v_5)$

$\therefore \varepsilon_{15} = \frac{u_5 - v_5}{2L} = \frac{T_{15}}{AE}$ $\quad \therefore T_{15} = \frac{u_5 - v_5}{2L}AE$

$\varepsilon_{25} = \frac{3u_5 - \sqrt{3}v_5}{4L} = \frac{T_{25}}{AE}$ $\quad \therefore T_{25} = \frac{3u_5 - \sqrt{3}v_5}{4L}AE$

$$\varepsilon_{35} = \frac{u_5}{L} = \frac{T_{35}}{AE} \qquad \therefore T_{35} = \frac{u_5}{L}AE$$

$$\varepsilon_{45} = \frac{u_5+v_5}{2L} = \frac{T_{45}}{AE} \qquad \therefore T_{45} = \frac{u_5+v_5}{2L}AE$$



$$(\rightarrow)\quad \frac{\sqrt{2}}{2}T_{15} + \frac{\sqrt{3}}{2}T_{25} + T_{35} + \frac{\sqrt{2}}{2}T_{45} = 0$$

$$(\uparrow)\quad \frac{\sqrt{2}}{2}T_{15} + \frac{1}{2}T_{25} = \frac{\sqrt{2}}{2}T_{45} + W$$

$$\therefore \frac{\sqrt{2}(u_5-v_5)}{4L}AE + \frac{3\sqrt{3}u_5 - 3v_5}{8L}AE + \frac{u_5}{L}AE + \frac{\sqrt{2}(u_5+v_5)}{4L}AE = 0$$

$$\therefore \frac{3\sqrt{3} + 4\sqrt{2} + 8}{8}u_5 - \frac{3}{8}v_5 = 0$$

$$\therefore v_5 = \frac{3\sqrt{3} + 4\sqrt{2} + 8}{3}u_5$$

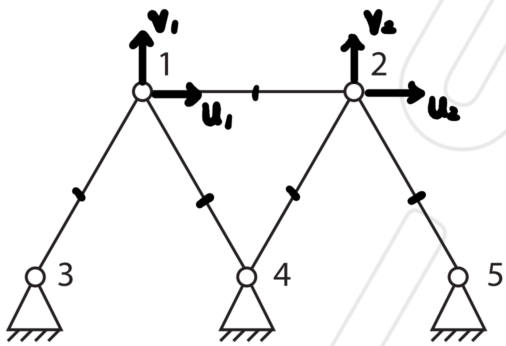
$$\therefore \frac{\sqrt{2}(u_5-v_5)}{4L}AE + \frac{3u_5 - \sqrt{3}v_5}{8L}AE = \frac{\sqrt{2}(u_5+v_5)}{4L}AE + W$$

$$\therefore (-\sqrt{3} - 4\sqrt{2})v_5 + 3u_5 = \frac{8WL}{AE}$$

$$\therefore u_5 = -0.1842\frac{WL}{AE} \qquad \therefore v_5 = -1.1575\frac{WL}{AE} \qquad \therefore W = \ldots$$

(**6.**) All the bars in the framework below have the same l, E, and A. The bar 1-3 (which has CTE α) is heated by ΔT°. Using the procedure outlined in the lectures for statically-indeterminate structures, solve for the joint displacements and show that the maximum strain in the framework is

$$\epsilon_{13} = \frac{4}{5}\alpha \Delta T$$



Answer $u_1 = \frac{3}{5}L\alpha\Delta T$, $v_1 = \frac{1}{\sqrt{3}}L\alpha\Delta T$, $u_2 = \frac{2}{5}L\alpha\Delta T$, $v_2 = 0$

$$M = b + r - 2j = 5 + 6 - 2\times 5 = 1$$

$\therefore$ statically indeterminate with 1 redundancy

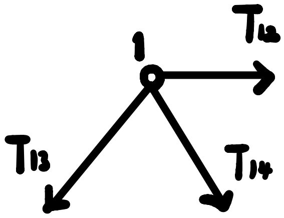
$$e_{13} = \frac{1}{2}u_1 + \frac{\sqrt{3}}{2}v_1 = \varepsilon_{13}L = \frac{4}{5}\alpha\Delta T L,$$

$$e_{12} = u_2 - u_1,$$

$$e_{14} = -\frac{1}{2}u_1 + \frac{\sqrt{3}}{2}v_1, \quad e_{24} = \frac{1}{2}u_2 + \frac{\sqrt{3}}{2}v_2$$

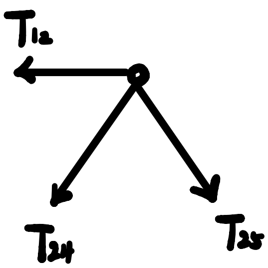
$$\therefore \varepsilon_{13} = \frac{e_{13}}{L} = \frac{T_{13}}{AE} + \alpha\Delta T = \frac{4}{5}\alpha\Delta T$$

$$\therefore \frac{T_{13}}{AE} = -\frac{1}{5}\alpha\Delta T \qquad \therefore T_{13} = -\frac{1}{5}\alpha\Delta T AE$$



$$(\uparrow)\quad \frac{\sqrt{3}}{2}T_{13} + \frac{\sqrt{3}}{2}T_{14} = 0 \qquad \therefore\ T_{14} = -T_{13}$$

$$(\rightarrow)\quad \frac{1}{2}T_{13} = \frac{1}{2}T_{14} + T_{12} \qquad \therefore\ T_{12} = T_{13}$$



$$(\uparrow)\quad \frac{\sqrt{3}}{2}T_{24} + \frac{\sqrt{3}}{2}T_{25} = 0 \qquad \therefore\ T_{24} = -T_{25}$$

$$(\rightarrow)\quad \frac{1}{2}T_{25} = \frac{1}{2}T_{24} + T_{12} \qquad \therefore\ T_{12} = T_{25}$$

$$\therefore\ T_{25} = T_{13} \qquad \therefore\ T_{24} = -T_{13}$$

$$\therefore\ e_{14} = -\frac{1}{2}u_1 + \frac{\sqrt{3}}{2}v_1 = \frac{T_{14}L}{AE} = \frac{1}{5}\alpha\Delta TL$$

$$e_{12} = u_2 - u_1 = -\frac{1}{5}\alpha\Delta TL$$

$$e_{24} = \frac{1}{2}u_2 + \frac{\sqrt{3}}{2}v_2 = \frac{1}{5}\alpha\Delta TL$$

$$\therefore\ u_1 = \frac{3}{5}\alpha\Delta TL, \quad v_1 = \frac{\sqrt{3}}{3}\alpha\Delta TL,$$

$$u_2 = \frac{2}{5}\alpha\Delta TL, \quad v_2 = 0$$